



Upgrades And Extras

When server loads increase, you'll need to go hardware shopping. Look for this:

Processor

If you need a streaming server, your tasks are likely to be multimedia heavy. Consider adding a second processor to fulfil client requests faster.

Memory

Heavy traffic can generate a lot of temporary data on the server, making performance take a hit. Minimise the need for disk paging by installing additional memory modules.

Storage

If your server is going to be used as a file server, consider getting an extra hard disk. Improve performance using a RAID 0 (striped) configuration, or increase reliability by using a RAID 1 (mirrored) configuration, or better still, get both performance and reliability by using a RAID 0+1.

Backup (Tape, DVD)

Get a tape drive or DVD-R and backup all your data regularly. While tape backup is a very cost-effective solution, data access is faster with DVD-Rs.

was the most underpowered sub-system on these servers. The ATi Rage XL adapter found onboard on most of them is more than sufficient, and also saves one valuable PCI slot.

The PCS and Acer Altos G310 were the only ones to ship with Geforce Mx440 and SiS Xabre AGP cards respectively.

The Power Supply: Handling The Load

Both the Wipro NetPower and the Acer Altos G520 offered high-wattage power supplies to feed the most demanding of loads. The 600 and 460 Watts offered by the Wipro and the Acer systems respectively are enough juice for bundled peripherals and more.

The PCS Siirus, Zenith's Data Server and Acer's Altos G310 offered 300-watt power supplies, which is barely adequate if there will be upgrades. Similarly, the HP ProLiant ML-110's 350-watt power supply is enough for the present hardware, but would need upgrading later.

The Wipro NetPower and Zenith's Data Server were the only ones that offered a power supply with an exhaust fan over the processor area for effective cooling.

What Comes Bundled

15-inch monitors, keyboards, mice, driver CDs, etc, were bundled with all the servers. Some, strangely enough, missed out on manuals. CD-ROM and floppy drives were common across all the servers, except for the Zenith, which offered a CD-RW—good for instant backups. Acer and HP had the best package content.

EasyBUILD, the customised server management system from Acer, was simple to use and understand. HP provided diagnostic tools. Zenith came bundled with Intel's server management tools, while both Wipro and PCS had tools that allowed for some server management.

Ventilation is of paramount importance for a server's optimum operation. Machines based on the latest Intel chips, in particular, need good airflow

Accessibility, Ventilation, And More

A good server is more than just a performance machine. Being switched on 24x7 demands a great deal of attention and care from their manufacturers so that downtime is at a minimum.

These machines need to be accessible—access to hardware such as hard drives, memory modules, etc, should be easy to do without the need for any tools. But from a security point of view, access to the internals should be restricted to system administrators. This is where locking facilities come in. A chassis intrusion detection system, for example, informs an administrator in the event of unauthorised access.

Ventilation is of paramount importance for optimum operation of a server. Machines based on the latest Intel chips, in particular, need good airflow.

As a general rule, air should enter from the front of the chassis and exit from the rear, with the help of fans. Various factors determine this flow—chassis design: the dimensions of the server case, the placement of the fans, etc. The internal wires from the power supply also need to be clamped together so they don't block air passage.

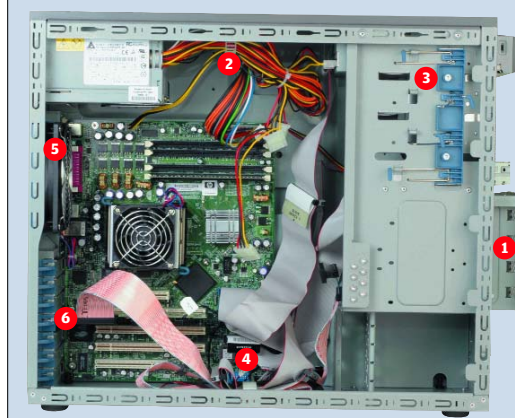
Wipro's Netpower had the roomiest cabinet of all—clutter was minimal. Ventilation is aided by a well-designed cabinet and is amply sup-



The Wipro NetPower 1120: Feature-loaded, expensive



What A Well-designed, Easy To Manage Server Must Have



1. Removable Hard disk bay
2. Properly tied power cables
3. Tool-free attachments for optical drives
4. Properly clamped IDE cables
5. Large fans to ensure proper ventilation
6. Easy to use tool-free clamps for expansion cards.